A. Summary of key equations and quantities

To provide a complete overview of the standard DEB model in one place, here follows a summary of key equations and quantities. The model dynamics covering the development of an organism from an egg to a fully mature adult are given by

$$\frac{dE}{dt} = \dot{p}_A - \dot{p}_C, \qquad \text{Reserve dynamics}$$

$$\frac{dL}{dt} = \frac{\dot{p}_G}{3L^2\,[E_G]}, \ \text{and} \qquad \text{Growth}$$

$$\frac{dE_H}{dt} = \begin{cases} \dot{p}_R, & \text{if } E_H < E_H^p \\ 0, & \text{if } E_H = E_H^p \end{cases}. \qquad \text{Maturation}$$

For these ordinary differential equations to be solvable, energy flows should be specified in terms of the state variables

$$\dot{p}_A = \begin{cases} 0, & \text{if } E_H < E_H^b \\ \{\dot{p}_{Am}\}\, f L^2, & \text{otherwise} \end{cases} \qquad \text{Assimilation}$$

$$\dot{p}_C = [E]\,\frac{\dot{v}\,[E_G]\,L^2 + [\dot{p}_M]\,L^3 + \{\dot{p}_T\}\,L^2}{[E_G] + \kappa\,[E]}, \qquad \text{Utilization}$$

$$\dot{p}_G = [E_G]\,\frac{\kappa\dot{v}\,[E]\,L^2 - [\dot{p}_M]\,L^3 - \{\dot{p}_T\}L^2}{[E_G] + \kappa\,[E]}, \ \text{and} \qquad \text{Growth}$$

$$\dot{p}_R = (1-\kappa)\,\dot{p}_C - \dot{k}_J E_H. \qquad \text{Maturation}$$

Important symbols appearing in the model equations are conveniently summarized in Table A.1.

To close the life cycle of an organism, an estimate of the reproductive output is still needed. The rate of continuous egg production, for example, is estimable using

$$\dot{R} = \frac{\kappa_R}{E_0} \times \begin{cases} 0, & \text{if } E_H < E_H^p \\ \dot{p}_R, & \text{if } E_H = E_H^p \end{cases}. \qquad \text{Egg production rate}$$

An alternative to continuous egg production is intermittent reproduction limited to a suitable window of opportunity called the reproductive season. Between two consecutive reproductive seasons, energy allocated to reproduction is stored in a buffer according to

$$\frac{dE_R}{dt} = \begin{cases} 0, & \text{if } E_H < E_H^p \\ \dot{p}_R, & \text{if } E_H = E_H^p \end{cases}. \qquad \text{Reproduction buffer}$$

Obtaining the rate of intermittent egg production from energy in the reproduction buffer requires defining species-specific buffer handling rules.

Although not strictly necessary from a mathematical perspective, in some applications (e.g., ecotoxicology), it is useful to explicitly write down the two maintenance flows as

$$\dot{p}_S = [\dot{p}_M]\,L^3 + \{\dot{p}_T\}\,L^2, \ \text{and} \qquad \text{Somatic maint.}$$

$$\dot{p}_J = \dot{k}_J E_H, \qquad \text{Maturity maint.}$$

which can then be used to slightly simplify the equations for other flows, i.e., utilization, growth, and maturation. The reason for doing such a simplification and emphasizing the role of maintenance flows is that some toxicants, e.g., xenobiotics, directly influence the value of maintenance-related parameters. Instead of having an environmental effect on the parameter values scattered across multiple model equations, it is usually a better practice to capture these changes in a single equation whenever possible.

One more equation consistently used in conjunction with the standard DEB model is the Arrhenius relationship. This relationship captures the effect of temperature on the metabolic rates of ectotherms

$$\dot{p}_*(T) = \dot{p}_*(T_{ref}) \exp\left(\frac{T_A}{T_{ref}} - \frac{T_A}{T}\right). \qquad \text{Arrhenius rel.}$$

When additionally the temperature tolerance range of an ectothermic organism needs to be accounted for, the Arrhenius relationship is extendable by multiplying its right-hand side by ratio $\gamma(T_{ref})/\gamma(T)$, where function $\gamma = \gamma(T)$ is given in Eq. (2).

Table A.1: Key quantities appearing in the standard DEB model.

Symbol	Description	Unit
$E(t)$	Energy in reserve	J
$L(t)$	Structural length	cm
$E_H(t)$	Level of maturity	J
$E_R(t)$	State of the reproduction buffer	J
E_0	Initial energy reserve of an egg	J
$f(t)$	Scaled functional response (see Section 6.2)	–
T	Body temperature	K
$\{\dot{p}_{Am}\}$	Maximum surface-area-specific assimilation rate	$\mathrm{J\,d^{-1}\,cm^{-2}}$
$\dot{v}$	Energy conductance	$\mathrm{cm\,d^{-1}}$
κ	Allocation fraction to soma	–
κ_R	Reproduction efficiency	–
$[\dot{p}_M]$	Volume-specific somatic maintenance cost	$\mathrm{J\,d^{-1}\,cm^{-3}}$
$\{\dot{p}_T\}$	Surface-area-specific somatic maintenance cost	$\mathrm{J\,d^{-1}\,cm^{-2}}$
$\dot{k}_J$	Maturity maintenance rate coefficient	$\mathrm{d^{-1}}$
$[E_G]$	Volume-specific cost of structure	$\mathrm{J\,cm^{-3}}$
E_H^b	Maturity at birth	J
E_H^p	Maturity at puberty	J
T_A	Arrhenius temperature	K
T_{ref}	Reference body temperature for parameter values	K

B. Embryonic development and the initial conditions

In DEB theory, an egg is assumed to initially contain only reserve received from the mother. By utilizing this reserve the embryo develops until becoming capable of feeding on its own. Under such conditions, as shown in Section 6.5, the triplet of the scaled state variables (e, l, e_H) at scaled time $\tau = 0$ is $\left(+\infty, 0, e_H^0\right)$, where $e_H^0 = (1-\kappa)\, g$. Because the embryo is not feeding on an outside food source, the scaled energy density is decreasing until it reaches value e_b at the moment of first feeding (i.e., birth in DEB terminology because the first feeding represents a major shift in the energy budget of any organism). The value of the scaled reserve density at birth is assumed to be known due to the maternal effect [110]. Namely, $e_b = f$, where f is food availability experienced by the mother. These considerations indicate that the scaled reserve density is a monotonically decreasing function of time (from $e = +\infty$ initially to $e = e_b$ at birth), thus allowing us to simplify the system of Eqs. (32) and (33) by using e as an independent variable instead of scaled time τ. We get an ordinary differential equation

$$\frac{dl}{de} = -\frac{l}{3e}\frac{e-l}{e+g}, \tag{B.1}$$

where we have taken into account that embryos do not feed ($f = 0$) and have negligible surface-area related somatic maintenance costs ($l_T = 0$).

Eq. (B.1) has an exact solution

$$l\,(e) = \frac{2g}{-2 + 2Cg^{4/3}\,(1 + e/g)^{1/3} + (1 + e/g)\,\Re\left[{}_2F_1\,(2/3, 1, 5/3, 1 + e/g)\right]}, \tag{B.2}$$

where C is an unknown integration constant, $\Re$ is the real part of a complex number, and ${}_2F_1(\cdot,\cdot,\cdot,\cdot)$ is a hypergeometric function. The correctness of this solution can be proven by inserting Eq. (B.2) back to Eq. (B.1). More importantly, we see that determining the integration constant C is equivalent to determining scaled length at birth, l_b, because $l_b = l\,(e_b)$, which can be solved for l_b when C is known and vice versa. Function $l\,(e)$ can be slightly simplified by substituting $x = 1 + e/g$, which runs from $x_0 = +\infty$ to $x_b = 1 + e_b/g$ during the course of embryonic development.

Constant C is constrained by Eq. (35) for the scaled maturity density. However, the problem is considerably simplified if we substitute $h = l^3 e_H/(1-\kappa)$, upon which maturity is tracked with equation

$$\frac{dh}{dx} = \frac{k}{g}\frac{l\,(x)}{x-1}h\,(x) - [l\,(x)]^3\,\frac{l\,(x)+g}{x}, \tag{B.3}$$

where $h(x_0) = 0$ and $h(x_b) = l_b^3 e_H^b/(1-\kappa)$—a known value because $l_b^3 e_H^b = E_H^b/([E_m]\,L_m^3)$. Eq. (B.3) is recognizable as a first-order linear differential equation of the form $\frac{dh}{dx} + P\,(x)\,h = Q\,(x)$ [180]. In our case, $P\,(x) = -\frac{k}{g}\frac{l(x)}{x-1}$ and $Q\,(x) = -\,[l\,(x)]^3\,\frac{l(x)+g}{x}$. The general solution is [180]

$$\begin{aligned} h\,(x) &= \exp\left(-\int_{x_b}^{x} P\,(x')\,dx'\right) \\ &\quad\times\left[h\,(x_b) + \int_{x_b}^{x} Q\,(x')\exp\left(\int_{x_b}^{x'} P\,(x'')\,dx''\right)dx'\right]. \end{aligned} \tag{B.4}$$

Fortunately, we do not need to solve this equation to find the value of constant C. Instead, it is sufficient to use the fact that $h(x_0) = 0$, yielding

$$h\,(x_b) = -\int_{x_b}^{x_0} Q\,(x')\exp\left(\int_{x_b}^{x'} P\,(x'')\,dx''\right)dx'. \tag{B.5}$$

From this condition, constant C is readily calculated using numerical integration. Once the value of C is known, the initial condition for simulating post-embryonic development with the standard DEB model is $\left(e_b, l_b(e_b), e_H^b\right)$.

If constant C is known, can it be used to calculate the initial energy reserve of an egg, E_0? The answer is affirmative, and the relationship to perform this calculation is

$$E_0 = \frac{[E_m]\, L_m^3}{\left(C - \frac{1}{4}\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{5}{3}\right)\right)^3}, \tag{B.6}$$

where $\Gamma(\cdot)$ is the gamma function. Proving Eq. (B.6) is quite technical and laborious, and hence omitted here. Fortunately, it is possible to test its correctness numerically against an equivalent relationship [110] (see also Section 2.6 in [7]). Given the numerical example in Fig. 3 with $\kappa = 0.8$, $E_H^b = 7.425\,\mathrm{J}$, and $k \approx 0.9333$ at $e_b = f = 0.8$, we obtain $C = 12.5497$. This value for C gives $l_b = 0.05075$ and $E_0 = 47.6\,\mathrm{J}$, which are precisely the expected results.